

$$A.1-) y' + \underbrace{\left(2 + \frac{1}{x}\right)}_{p(x)} y = \underbrace{e^{-2x}}_{q(x)}$$

$$y = e^{-\int p(x) dx} \cdot \left[c + \int q(x) e^{\int p(x) dx} dx \right] = e^{-\int \left(2 + \frac{1}{x}\right) dx} \cdot \left[c + \int e^{-2x} \cdot e^{\int \left(2 + \frac{1}{x}\right) dx} dx \right]$$

$$y = e^{-2x} \cdot e^{-\ln x} \cdot \left[c + \int e^{-2x} \cdot e^{2x} \cdot e^{\ln x} dx \right]$$

$$= e^{-2x} \cdot \left(\frac{1}{x}\right) \cdot \left[c + \int \cancel{x} dx \right] = \frac{e^{-2x}}{x} \left[c + \frac{x^2}{2} \right]$$

$$A.2-) (x^2+1) \cdot y' + 4xy = x, y(2)=1$$

$$y' + \underbrace{\left(\frac{4x}{x^2+1}\right)}_{p(x)} \cdot y = \underbrace{\frac{x}{x^2+1}}_{q(x)} ; y = e^{-\int p(x) dx} \cdot \left[c + \int q(x) e^{\int p(x) dx} dx \right]$$

$$y = e^{-\int \frac{4x}{x^2+1} dx} \cdot \left[c + \int \frac{x}{x^2+1} \cdot e^{\int \frac{4x}{x^2+1} dx} dx \right]$$

$$\Rightarrow x^2+1 = u \text{ olsun } 2x dx = du$$

$$y = e^{-2 \int \frac{du}{u}} \cdot \left[c + \int \frac{x}{x^2+1} \cdot e^{2 \int \frac{du}{u}} dx \right]$$

$$= e^{-2 \cdot \ln u} \cdot \left[c + \int \frac{x}{x^2+1} \cdot e^{2 \ln u} dx \right]$$

$$= \frac{1}{(x^2+1)^2} \cdot \left[c + \int \frac{x}{x^2+1} \cdot (x^2+1)^2 dx \right]$$

$$= \frac{1}{(x^2+1)^2} \left[c + \int x(x^2+1) dx \right]$$

$$= \frac{1}{(x^2+1)^2} \left[c + \frac{x^4}{4} + \frac{x^2}{2} \right]$$